

HealthBridge — Recommended Firebase Tree Structure

This document defines the recommended Firebase Realtime Database structure for the HealthBridge project before adding Wear OS watch integration. Goals of this structure: - clean and scalable architecture - separation of GPS, watch, and device data - easier debugging - easier synchronization - future support for caregivers, alerts, analytics, and additional sensors

Recommended Firebase Tree

```
groups
├── family_001
│   ├── members
│   │   ├── alain
│   │   │   ├── profile
│   │   │   │   └── name : "Alain"
│   │   │   ├── latest
│   │   │   │   ├── timestamp
│   │   │   │   └── location
│   │   │   │       ├── lat
│   │   │   │       ├── lng
│   │   │   │       └── altitude
│   │   │   ├── watch
│   │   │   │   ├── heartRate
│   │   │   │   ├── spo2
│   │   │   │   ├── steps
│   │   │   │   └── battery
│   │   │   └── device
│   │   │       ├── phoneBattery
│   │   │       └── lastSeen
│   │   └── mary
│   │       └── SAME STRUCTURE
```

1. Profile Node

Contains static information about the member.

Example:
profile/name

This area can later include:

- role
 - age
 - emergency contact
 - permissions
-

2. Latest Node

Contains the most recent real-time information.

The latest node is updated continuously by the phone and the watch.

3. Location Node

Contains GPS information.

Fields:

- lat
- lng
- altitude

This keeps GPS data isolated from watch telemetry.

4. Watch Node

Contains Wear OS watch data.

Fields:

- heartRate
- spo2
- steps
- battery

This modular structure is important for future expansion.

5. Device Node

Contains phone status information.

Fields:

- phoneBattery
- lastSeen

This allows caregiver alerts such as:

- phone offline
 - low battery
 - no updates received
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6. Important Architecture Rule

Each phone publishes ONLY its own member node.

Example:

- Alain phone writes ONLY to members/alain
- Mary phone writes ONLY to members/mary

No phone should write another member.

7. Recommended Cleanup

Old temporary test fields should eventually disappear.

Examples:

- latitude
- longitude
- status
- fakeGps

All production GPS should be stored only inside:

